



Predicting uncertainty in vision-based satellite pose estimation using deep evidential regression

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ABSTRACT

In this work, we demonstrate the use of evidential regression to quantify uncertainty in satellite pose estimation within vision-based deep-learning models trained on images of the Tango satellite. Our approach augments existing deep-learning models with uncertainty prediction capability using evidential regression. We show that the resulting uncertainty is well correlated with prediction errors, while maintaining comparable pointwise-prediction accuracies to existing models. Leveraging evidential regression to predict epistemic and aleatoric uncertainties, we demonstrate a median 0.996 correlation between network prediction error and these uncertainty estimates on keypoint coordinates within synthetic images. Additionally, we demonstrate that predicting on out-of-distribution images as well as increasing image noise directly amplifies uncertainty in pose estimation. By validating our framework's ability to quantify uncertainty, we enable robust decision-making in risky satellite operations.

1. Introduction

Predicting the position and orientation of a satellite from images is a challenging task that is crucial for a range of applications, from satellite navigation [1–3], docking [4,2], to debris removal [5,6]. Machine learning approaches to image-based pose prediction are required for autonomous real-time systems where humans not only stay out-of-the-loop for decision making, but also where expensive equipment is not available. Smaller and cheaper satellites may not be able to use many refined sensors such as LIDAR to track oncoming satellites, where monocular images may already be available. Recent advances in deep learning have significantly improved the accuracy of pose estimation for uncooperative satellites from images [7–9]. These deep-learning approaches, however, are difficult to trust, as they often produce similar predictions on highly corrupted images as they do on noise-free ones, without signaling that the corrupted input may lead to an incorrect pose estimate. These models fail to provide insights into the reliability of their predictions, demonstrating remarkably worse performance on out-of-distribution datasets. Encompassing these untrustworthy machine learning models inside an autonomous controls law may lead to disastrous decision-making. Our work addresses this gap by applying a method to quantify uncertainty in deep learning pre-

dictions for satellite pose estimation. The ability for the network to indicate its confidence of a prediction enables the control law to not trust the network, enhancing safe and reliable autonomous decision-making.

Existing works that address uncertainty in the pose estimation framework [10–12] focus on enhancing the accuracy of pose prediction or trajectory tracking, while failing to validate their uncertainty on actual pose prediction error. Our work uses a different approach by predicting distributions of important key points of the satellite using evidential regression [13]. This methodology enables aleatoric and epistemic uncertainty measures while also directly indicating the model's confidence in the pose estimate.

We use a compilation of synthetic computer-generated satellite images known as the SPEED+ dataset [14] to train and validate our predictive network. The dataset features two collections of real images of miniature staged satellites to verify our network on unseen out-of-distribution data. Our approach leverages a deep neural network architecture designed to predict landmark coordinates from a given image. These coordinates are then used to estimate both the position and orientation of the pictured satellite. By incorporating an evidential regression framework, we are able to also model two types of uncertainties: epistemic, which captures model uncertainty due to limited training data, and aleatoric, which accounts for inherent noise in the observations.

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Learning these two types of uncertainties enables a more robust understanding of prediction reliability.

In our analysis with other pose-predictive networks, we demonstrate a strong correlation of 0.996 between predictive uncertainty and prediction error in keypoint coordinates on synthetic images. This validation of our uncertainty quantification framework emphasizes its effectiveness in identifying when the model is likely to be wrong. Moreover, our results show that incorporating uncertainty estimates does not compromise the accuracy of satellite pose predictions. On the contrary, we achieve prediction errors that are 13% and 8% smaller for translation and orientation predictions compared to a benchmark network on synthetic images. We also display comparable accuracies on out-of-distribution datasets while providing valuable uncertainty estimates. We see a median 0.9 correlation between translation errors and predicted uncertainty for both out-of-distribution datasets. Similarly, on both out-of-distribution datasets we see a median 0.6 correlation between orientation errors and predicted uncertainty.

Our main contribution includes augmenting deep-learning architectures with uncertainty quantification capabilities for satellite pose estimation without sacrificing performance accuracies. We demonstrate increased network uncertainty on out-of-distribution data and noisy images to further validate the evidential regression framework. Finally, we utilize uncertainty-aware perspective-n-point methods to predict the pose of the satellite using covariance information from the network's prediction.

Reliable uncertainty estimates can enhance decision-making processes by highlighting when predictions should be trusted and when caution is warranted. By applying a robust framework for predicting and understanding uncertainties, we enable safer and more reliable satellite operations. In Section 2, we cover the background on deep-learning satellite pose estimation, previous works, and the multiple datasets. Section 3 explains the regression method we use to quantify uncertainty. Section 4 details the methodology of our proposed networks, and finally, Section 5 demonstrates the performance of the proposed network with numerical results.

2. Pose estimation

In this section, we provide a background on deep-learning methods for pose estimation using monocular images, investigate the relevant previous works in the field, and detail the datasets used in this work.

2.1. Background

Pauly et al. published a survey on deep-learning approaches to satellite pose estimation [15] that categorizes the current methodologies into two categories, hybrid [16,17,8] and direct end-to-end methods [18,19,9]. Hybrid approaches use a combination of classical pose estimation techniques and deep-learning strategies while the direct end-to-end methods rely entirely on deep-learning architectures to predict a satellite's pose. Given an image of a spacecraft, a direct end-to-end approach uses convolutional neural networks to predict the position and orientation of a satellite directly. A hybrid technique is largely composed of bounding box estimation, keypoint regression, and finally, pose estimation. In this work, we focus on developing a hybrid approach and compare our results to both a hybrid method, KRN [8], and an end-to-end method, SPNv2 [9].

The first step in a hybrid approach is bounding box estimation which stems from an object detection task. The goal of the object detection network (ODN) is to learn the pixel coordinates of a bounding box of an object given an input image. For spacecraft object detection, the neural network will have to be trained on a specific satellite to be able to identify and isolate the satellite in an image. After a bounding box of the spacecraft is acquired, a separate deep-learning architecture is used to predict the pixel coordinates of predefined keypoints of a spacecraft. These keypoints are chosen offline from identifiable points of interest

that can be easily identified from an image. Once a network is trained to identify the 2D pixel coordinates, the pose of the spacecraft needs to be estimated from these points. A popular algorithm for this transformation is the efficient perspective-n-point (EPnP) algorithm [20]. The hybrid approach takes these three tasks into one pipeline that predicts the pose of a satellite.

2.2. Previous works

To the best of the authors' knowledge, Sharma, Beierle, and D'Amico [7] were the first to propose deep learning for spacecraft pose estimation by using a convolutional neural network for classification in pre-defined pose classes. Later Sharma and D'Amico [21] developed the spacecraft pose network (SPN) that used a combination of classification and regression techniques for pose estimation. The SPN network outputs a 2D bounding box which is given to two subsections of the network that solve a classification and regression problem respectively. The first sub-network finds the pre-specified pose classes that are the closest to the pose in the given image while the second sub-network assigns weights to those top classes. The resulting pose is a linear combination using the closest classes and their associated weights. The SPN network is a direct end-to-end approach that does not rely on keypoints or a PnP method.

The keypoint regression network (KRN) was introduced in [8] as a solution to estimating the pose of a satellite using keypoints. Their ODN is comprised of the MobileNetv2 and the YOLOv3 LITE networks to predict the bounding box of the satellite given an uncropped image. Once the image is cropped using the bounding box, the KRN, comprised of the MobileNetv2, YOLOv2 LITE, and additional convolutional layers, is used to predict the locations of the 2D pixel locations of the keypoints of the satellite. Using the known keypoints and the 2D pixel predictions, the EPnP algorithm is able to predict the rotation quaternion and the translation vector for the pose of the satellite. The KRN network was one of the first examples of using deep learning in a hybrid approach.

The spacecraft pose network v2 (SPNv2) was introduced in [9] as an upgraded version of the SPN [21]. The larger network, SPNv2, is comprised of a shared feature encoder and multiple prediction heads. For the common feature encoder, an EfficientNet and a bi-direction feature pyramid network are combined to learn the shared information that is then passed to the multiple prediction heads. Each prediction head has a unique structure that is chosen for its desired task. The SPNv2 predicts the pose class, the bounding box, the translation vector, a rotation matrix, a heatmap of the predicted locations of the keypoints, and a segmentation map of the foreground of the satellite image. The combination of predicting multiple related tasks enables the network to generalize well to unseen data. In this work, we compare our introduced network to the SPNv2 results in [9].

Since the two spacecraft pose estimation challenges in 2019 [22] and 2021 [23], many other notable methods have been developed. This includes a hybrid approach by [24] who jointly regresses keypoint locations and depth while introducing an unsupervised domain adaptation to achieve high pose estimation accuracies on out-of-distribution data. The network proposed by [16] employs a self-training approach that iteratively optimizes keypoints using heatmaps and temporal information to enhance training accuracy. Another unsupervised domain adaptation technique by [25] incorporates geometrical constraints and segmentation masks, outperforming existing methods on out-of-distribution data. However, these methods often come with significant computational costs. To address this, lightweight networks from [26] have been designed for monocular-image based pose estimation on actual hardware using tensor processing units. Similarly, [19] introduced a highly compact network that is orders of magnitude smaller than its counterparts, achieving only slightly worse accuracy while significantly enhancing on-board capabilities.

All of the above-mentioned networks, however, fail to identify when the network is not confident in its prediction. Uncertainty prediction in satellite pose estimation was previously studied in [12]. After recov-

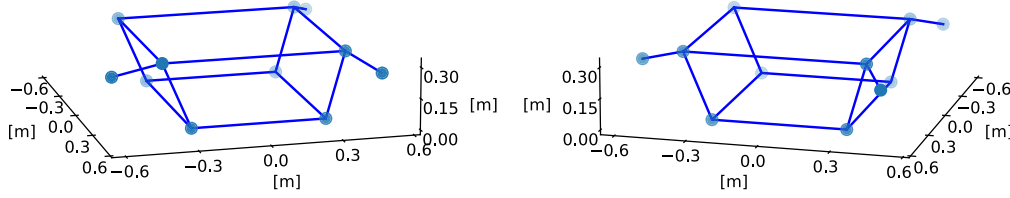


Fig. 1. Wireframe of Tango satellite from two orientations.

er the keypoints of a satellite and training an ODN, Li et al. predict individual bounding boxes for each keypoint and predict the probability of finding each keypoint in the bounding boxes. Using this prediction of confidence for each keypoint, they develop two methods to reduce the error of pose estimation. The Top-K method selects the K keypoints with the highest predicted confidence and uses a PnP method to predict the pose using those K keypoints. This method was shown to reduce the error in pose estimation by a reasonable margin by simply eliminating keypoints that introduce unnecessary error. The other method, uncertainty threshold selection, uses all keypoints with confidence greater than a specified level. Both of these methods filter out keypoints with large uncertainty in predictions to provide the PnP method only with accurate keypoint predictions. This enables the pose estimation to have a higher accuracy compared to methods without uncertainty predictions. Predicting probabilities of keypoints in bounding boxes, however, does not enable uncertainty quantification of the pose estimation.

To estimate the pose uncertainty, the proposed approach by [10] uses covariance estimates of heatmap predictions of a keypoint regression network. The covariance of each keypoint's heatmap is extracted and fitted to a Gaussian distribution where the CEPnP [27] method is used to predict pose. The covariance information from the heatmaps are also passed to a navigational filter to more accurately estimate a satellite's trajectory. While the framework predicts the uncertainty of a pose prediction, the results do not demonstrate the accuracy of the uncertainty predictions or test the uncertainty predictions on the network's error.

Finally, the introduced method by [11] also fits a Gaussian distribution over covariance predictions of keypoint coordinates to estimate the uncertainty of the satellite pose estimate. The keypoint location covariance is found, however, by re-projecting previously found 3D keypoints into the 2D image with various lighting conditions. The covariance with respect to different lighting conditions is also passed to a navigational filter to more accurately estimate a satellite's trajectory. Similarly to [10], while the uncertainty of the pose estimate is found, the predicted uncertainty is not validated or compared to the error of their pose estimation.

To the best of the authors' knowledge, all previous works that aim to quantify the uncertainty of pose estimation focus on improving pose estimation accuracy or implementing a more accurate filtering method. The previous works have not validated their uncertainty predictions against prediction errors. In this work, we validate our uncertainty predictions against pose estimation error and provide uncertainty using non-Gaussian distributions.

In the Section 3, we introduce a method to assign probability distributions to each keypoint to enable taking samples of the translation and orientation of the satellite. This contribution enables us to quantify the uncertainty directly on the pose of the satellite. By measuring the spread of the translation and orientation samples, we are able to quantify the confidence of the network in its prediction.

2.3. Datasets

To train our proposed method, the SPEED+ dataset [14] was chosen for its synthetic and hardware-in-the-loop datasets. The SPEED+ repository consists of three datasets of artificial images of the Tango satellite from the PRISMA mission [28]. The SPEED+ dataset is an extension of

the earlier SPEED dataset [29] and used for the Satellite Pose Estimation Competition in 2021 [23]. Since the SPNV2 uses the same dataset to train, the keypoints of the Tango satellite are available publicly from the SPNV2 Github repository¹ and are shown along with a wireframe of the satellite in Fig. 1.

The SPEED+ dataset contains nearly 60,000 synthetic images created using a camera emulator from the Stanford's Space Rendezvous Laboratory (SLAB). These synthetic images were split 80:20 to training and testing datasets. The other two datasets are images created from the Testbed for Rendezvous and Optical Navigation (TRON) from SLAB. The testbed uses a model of the Tango satellite and creates unique lighting scenarios from a *lightbox* for one dataset and a *sunlamp* for another. These datasets are useful for testing pose estimation networks on out-of-distribution data that may more accurately represent real-world scenarios.

3. Uncertainty measures and evidential regression

In this section, we will describe the methodology of predicting uncertainty in regression techniques. We aim to approximate a function $f(x) : \mathcal{X} \rightarrow \mathcal{Y}$ using a set of data by learning model parameters θ . To handle sources of uncertainty during regression, we model a function evaluation $y \in \mathcal{Y}$ as a Gaussian distribution whose parameters (m , σ^2) are functions of the input $x \in \mathcal{X}$

$$f(x) \sim \mathcal{N}(m(x; \theta), \sigma^2(x; \theta)), \quad (1)$$

where θ are the parameters governing the learned models $m(x)$ and $\sigma^2(x)$. Consider a set of data $D = \{(x_i, y_i); i = 1, \dots, N\}$ such that we seek a posterior over θ given the data D and prior information I

$$p(\theta | D, I) = \frac{1}{p(y | x, I)} p(y | \theta, x, I) p(\theta | x, I), \quad (2)$$

such that y is a vector of all outputs, x is a vector of all inputs and I is auxiliary information that helps define the prior. The distribution $p(\theta | D)$ contains the best-fit parameters and represents the epistemic uncertainty of the Gaussian model. As the number of data points increases, $m(x)$ and $\sigma^2(x)$ will converge to specific functions, and the epistemic uncertainty converges to 0. Once θ is given, $m(x)$ and $\sigma^2(x)$ are deterministically specified. Thus, $\sigma^2(x)$ measures the aleatoric uncertainty representing the "randomness" in the data while the uncertainty in θ represents the epistemic uncertainty by the lack of data.

The regression method that we focus on is deemed deep evidential regression and takes a hierarchical approach to the prediction distribution [13]. Our description below follows the same reference. While the mean $m(x)$ and variance $\sigma^2(x)$ are both functions of the input x , we drop the function notation for readability. Evidential regression first assigns a prior on m and σ^2 , rather than specifying them as parametric functions. This prior is in some parametric family defined by parameters η so that

$$p(m, \sigma^2 | D, \eta) = \frac{1}{p(y | x, \eta)} p(y | m, \sigma^2) p(m, \sigma^2 | \eta) \quad (3)$$

¹ The SPNV2 network and data augmentations are available at <https://github.com/tpark94/spnv2>.

where we replaced the information I in Equation (2) by the parameters of the prior distribution, η . We may choose the prior to be separable using a normal distribution and an inverse-gamma distribution

$$m \sim \mathcal{N}(\gamma, \nu^{-1}\sigma^2) \quad \sigma^2 \sim \Gamma^{-1}(\alpha, \beta) \quad (4)$$

parameterized by $\eta = (\gamma, \nu, \alpha, \beta)$ where $\gamma(x)$, $\nu(x)$, $\alpha(x)$, and $\beta(x)$ are also functions of the input x . These distributions are then used to construct a Normal Inverse-Gaussian prior

$$p(m, \sigma^2 | x, \eta) = p(m | \sigma^2, x, \eta) p(\sigma^2 | x, \eta) \quad (5)$$

$$\begin{aligned} &= \sqrt{\nu} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\nu \frac{(m-\gamma)^2}{\sigma^2}\right) \\ &\times \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left(-\frac{\beta}{\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi}} \frac{\sqrt{\nu}\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha+\frac{3}{2}} \exp\left(-\frac{1}{2\sigma^2}(\nu(m-\gamma)^2 + 2\beta)\right) \end{aligned} \quad (6)$$

where the additional information $\eta(x; \theta) = (\gamma(x; \theta), \nu(x; \theta), \alpha(x; \theta), \beta(x; \theta))$ are the hyperparameters of the prior with constraints $\gamma \in \mathbb{R}$, $\nu > 0$, $\alpha > 1$, $\beta > 0$. Note that these parameters are functions of the input x and also parameterized by some θ .

The likelihood $p(y | m(x), \sigma^2(x))$ assumes independent observations and is taken directly from (1)

$$L(D) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} (\sigma^2(x_i))^{-1/2} \exp\left(-\frac{1}{2} \frac{(f_i - m(x_i))^2}{\sigma^2(x_i)}\right) \quad (8)$$

We will choose parameters θ by maximizing the marginal likelihood $p(y | x, \theta)$. Under this distribution the evidence is a location-scale t-distribution where (assuming independent data and no repeated inputs)

$$y_i | x_i, \gamma, \nu, \alpha, \beta \sim \text{lst}\left(y(x_i), \frac{\beta(x_i)(1 + \nu(x_i))}{\alpha(x_i)\nu(x_i)}, 2\alpha(x_i)\right) \quad (9)$$

and the log marginal likelihood of each data-point is

$$\begin{aligned} &\log p(y_i | \gamma, \nu, \alpha, \beta) \\ &= \log \frac{\Gamma(\alpha + 1/2)}{\Gamma(\alpha)} + \frac{1}{2} \log \frac{\nu}{\pi} + \alpha \log 2\beta(1 + \nu) \\ &\quad - (\alpha + 1/2) \log (2\beta(1 + \nu) + \nu(y_i - \gamma)^2) \end{aligned} \quad (10)$$

where it is understood that the right hand side has each of the four parameters evaluated at x_i .

In deep evidential regression [13], we find θ by maximizing Equation (10) plus a regularization term. The regularization term helps to decrease confidence on incorrect predictions

$$\mathcal{R}_i = |y_i - \gamma| \cdot (2\nu + \alpha). \quad (11)$$

The total loss is minimized by a weighted sum of the log marginal likelihood and the regularization term

$$\text{Loss}(\theta) = \sum_i^N -\log p(y_i | \gamma, \nu, \alpha, \beta) + \lambda \mathcal{R}_i \quad (12)$$

where λ is a chosen weight to balance regularization.

Once the hyperparameters are found, we make predictions directly using (9). The prediction at a new point is simply the expectation of the mean, $m(x)$. The aleatoric uncertainty is the expectation of the variance, $\sigma^2(x)$, and the epistemic uncertainty can be considered the variance of the mean, $m(x)$. The total uncertainty is the variance of the location-scale t-distribution (9), which is the sum of the aleatoric and epistemic uncertainties

$$\begin{aligned} &\underbrace{\mathbb{E}[m(x)] = \gamma(x)}_{\text{Prediction}}, \quad \underbrace{\text{Var}[f(x)] = \frac{\beta(x)(1 + \nu(x))}{\nu(x)(\alpha(x) - 1)}}_{\text{Total Uncertainty}}, \\ &\underbrace{\mathbb{E}[\sigma^2(x)] = \frac{\beta(x)}{\alpha(x) - 1}}_{\text{Aleatoric Uncertainty}}, \quad \underbrace{\text{Var}[m(x)] = \frac{\beta(x)}{\nu(x)(\alpha(x) - 1)}}_{\text{Epistemic Uncertainty}}. \end{aligned} \quad (13)$$

The predictions of the mean and uncertainties are all parameterized by the hyperparameters $\eta(x)$ of the prior distribution. A neural network is constructed to output the four hyperparameters $\eta(x)$.

4. Evidential keypoint regression network (EKRN)

In this work, we adapt the evidential regression framework [13] to new network architecture and enable uncertainty-aware PnP methods on satellite pose estimation. By sampling the predicted keypoint distributions, we also provide confidence bounds on a predicted pose that captures both aleatoric and epistemic uncertainty. We demonstrate that providing these uncertainty estimates increases satellite pose estimation accuracy and also verify the uncertainty estimates against network error.

We take a hybrid approach for estimating the satellite pose, using keypoint regression and pose estimation with the EPnP [20] and the direct least-squares with uncertainty (DLSU) algorithm [30]. Unlike previous works, we forgo the ODN and instead pass randomly cropped and shifted images of the satellite to the keypoint regression network. Using the labels of the bounding boxes, we imitate an incorrect ODN to robustify the keypoint regression network. The random cropping and shifting of the bounding boxes is taken from the KRN network implementation² [8]. Many ODNs for the Tango satellite have been created and may be inserted into this pipeline [8, 26, 31].

The keypoint regression network is constructed to predict the locations of pre-defined keypoint coordinates of the Tango satellite. Unlike previous works, the evidential regression network instead predicts hyperparameters $\eta(x)$ of an output distribution (9) that represents the probability of finding a keypoint at a pixel location. For pose prediction, we simply use the γ -prediction hyperparameter as the pixel locations of each of the keypoints and pass them through the EPnP algorithm to predict pose. For uncertainty estimates in pose, we sample the output distribution (9) of each keypoint and pass the samples through the cheap EPnP algorithm to obtain samples of the resulting pose distribution. Specifically, the marginal likelihood (9) is sampled for each of the keypoints while the epistemic and aleatoric uncertainties of the keypoint distributions can be found using (13). The samples of the satellite's pose allow us to find empirical standard deviations of each translation axis (X-Y-Z position) and orientation axis (Z-Y-X rotation) and are considered a measure of the uncertainty of the network's prediction. If a more general metric of the pose's uncertainty is required, we can find an aggregate uncertainty measure for translation and orientation separately. Using the pose samples, we can find the average distance of each sample from its sample mean for both translation and orientation to represent the spread of the translation predictions and orientation predictions. This average sample distance can be considered the uncertainty of the translation and orientation predictions respectively.

While the cheap EPnP algorithm is used to generate pose samples for uncertainty quantification, we also use the DLSU algorithm to predict the most likely pose given the 2D keypoint predictions and their uncertainties. Unlike most PnP methods, the DLSU algorithm is able to use the 2D uncertainties to find the pose by maximizing its log-likelihood. Other algorithms such as the CEPPnP [27], MLPnP [32], and EPnPU [30] are also able to handle 2D feature uncertainties, but the DLSU algorithm

² The KRN and SPN networks are available at <https://github.com/tpark94/speedplusbaseline>.

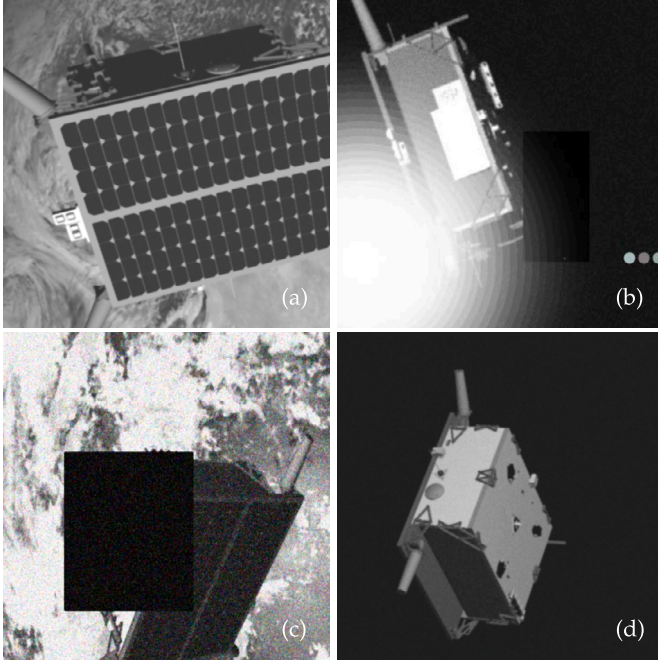


Fig. 2. Four example training images containing data augmentations from the synthetic SPEED+ dataset.

was compared and chosen in this work for its higher estimation accuracy on the results in Section 5. See Appendix A for further comparisons between PnP methods.

To improve the generalization of the trained networks, we apply random data augmentations to the training set of the synthetic images. The data augmentations are modified from the SPNv2 augmentations [9] using the Albumentations [33] package. We applied brightness and contrast edits while applying rectangular dropout sections, artificial sun flares, motion, glass and median blurs, Gaussian and ISO noise, and rotation and flipping. Each of the augmentations are randomly applied for each training batch of data. Compared to the SPNv2 modifications, we added random 90 degree rotations and horizontal and vertical flipping of the images. We also decreased the source size of the sun flare to half of the SPNv2 implementation to 200 pixels and decreased the maximum number of coarse dropout sections per image to 4. Similarly to the KRN implementation [8], we add random zoom and shift to the bounding box of the satellite to imitate a non-perfect bounding box predictor. Many images have keypoints cropped out or contain a non-centered satellite. Four example images of the training data after the augmentations can be seen in Fig. 2. Image (a) displays a zoomed-in picture with many keypoints out of view and pictures the Earth surface in the background. Image (b) shows a coarse dropout and a sun flare in the lower left with three sun flare effects in the lower right. Image (c) displays the random

Table 1
Training Parameters for the EKRn-HD.

Parameter	Value
Batch size	60
Epochs	200
Learning Rate	0.001
Weight Decay	0.01
Model Parameters	50,219,144
Augmentation Probability	0.5
Regularization Weight - λ	0.0001

noise with a large dropout section. Finally, image (d) is an unaltered image with black space in the background.

The first network we construct is the Evidential Keypoint Regression Network (EKRN) in direct comparison with the KRN network [8]. The architecture of the EKRN follows directly from the KRN network where the only difference between the two networks is the final output convolutional layers. The KRN outputs 22 values, the x- and y-pixel coordinates of the 11 keypoints. The EKRN duplicates the last convolutional layer into 4 separate layers sharing the same output of the YOLOv2 LITE. Each of the 4 layers outputs 22 values, corresponding to the γ , ν , α , and β for the x- and y-positions of the 11 keypoints. The positive constraints for ν , α , and β listed in Section 3 are enforced using a softplus layer. Diagrams of the KRN and EKRN are shown in Fig. 3.

To compare with the much larger network, SPNv2, we construct a comparable sized network using the Efficient Net B5, a modified YOLOv2 LITE, and additional convolutional layers. We deem the larger network as the EKRN high-definition (EKRN-HD) since the input image of the network is a 456px by 456px image, compared to the smaller input size of the KRN and EKRN networks of 224px by 224px. The EKRN-HD network also uses the larger EfficientNet B5 instead of the MobileNetv2. From the EKRN-HD network, we calculate the pose from the keypoint predictions in three different methods. The first method uses all 11 keypoint γ -predictions in the EPnP algorithm. The second uses the Top K strategy from [12] where we predict the pose using the top 9 keypoints with the least uncertainty and the EPnP algorithm. Since the x- and y-coordinates have a predicted total uncertainty (13), we add them together and sort to find the 9 most confident keypoints. The sum of the x- and y-coordinate uncertainties is equivalent to the trace of the joint x-y covariance matrix of its t-distribution. The more accurate keypoint predictions in the EPnP algorithm should improve the pose estimation accuracy. The third method uses the DLSU algorithm from [30] instead of EPnP which can use the network's covariance predictions to calculate a more accurate pose.

Training parameters for the EKRN-HD can be seen in Table 1. The total training time for the EKRN-HD was 25.6 hours using a NVIDIA H100 core GPU. The network uses the AdamW optimizer [34] with 60 images in each batch over 200 epochs. To find the best regularization weight λ , we trained three networks with λ -values of 10^{-3} , 10^{-4} , and 10^{-5} . The λ -value that was chosen (10^{-4}) corresponds to the network with the best prediction results.

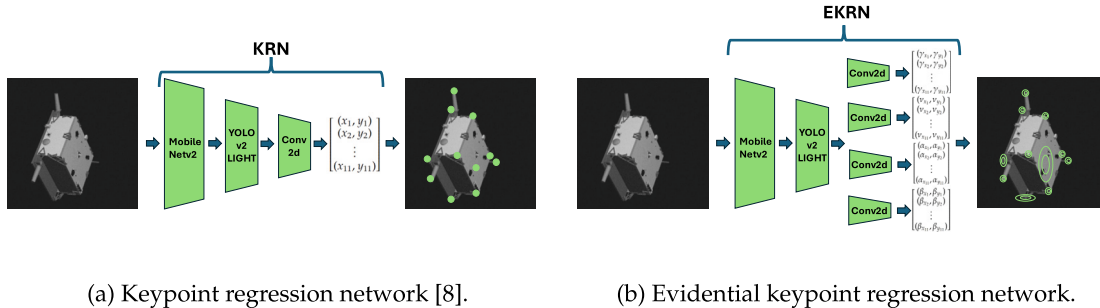


Fig. 3. Diagrams of the architecture of the KRN and EKRN are shown. The evidential network outputs four times the number of outputs of the KRN but provides hyperparameters for the hierarchical distributions instead of pixel locations of the keypoints.

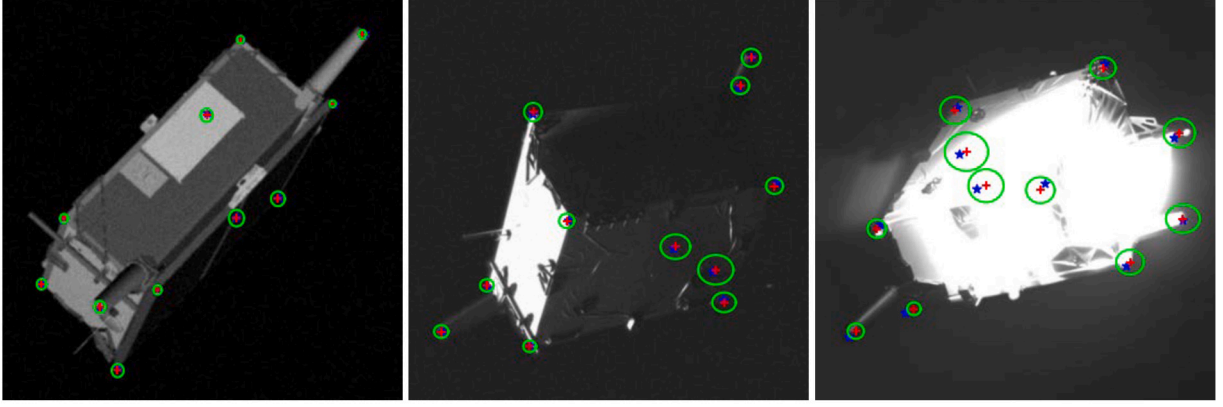


Fig. 4. The keypoints of synthetic, lightbox, and sunlamp test images respectively are predicted using the EKRN-HD with 99 percent quantiles shown in ellipses. The blue stars are the true keypoints while the red crosses are the γ -mean predictions. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 2

Results for average translation error $\bar{e}_T$ in meters and rotation error $\bar{e}_R$ in degrees on different testing datasets.

	Millions of Params.	Synthetic		Sunlamp		Lightbox	
		$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]
KRN [8]	6	0.361	6.52	2.01	53.48	1.31	31.36
EKRN	9	0.106	3.00	2.19	65.90	1.17	34.08
EKRN-HD	50	0.032	0.830	0.330	15.03	0.247	8.81
EKRN-HD-9	50	0.027	0.815	0.298	14.70	0.215	8.51
EKRN-HD*	50	0.034	0.750	0.271	13.95	0.200	8.01
SPNv2 [9]	52	0.031	0.885	0.230	10.37	0.216	7.98

Three example EKRN-HD predictions with test images from the synthetic, lightbox, and sunlamp datasets can be seen in Fig. 4 respectively. The true keypoint locations are denoted using the blue stars, and the red crosses are the γ -mean predictions of the EKRN-HD. The green ellipses represent the 99 percent quantiles of the keypoint distributions. Since the x- and y-coordinate of each keypoint is predicted independently, the semi-major and semi-minor axes of each ellipse are found using the 99 percent quantiles calculated from the t-distributions (9) of each coordinate-keypoint prediction. Notice that keypoints in clear view are shown with great confidence while keypoints on the hidden side of the satellite are shown with greater uncertainty. Other keypoints that are hard to identify from the bright sunlamp image are also given greater uncertainty ellipses.

5. Results

In Table 2, we compare six networks in translation and rotation error metrics for the testing synthetic, sunlamp, and lightbox data. Since the published KRN results in [8] are not trained on the augmented images in Section 4, we retrain the KRN on the augmented dataset for a fair comparison in Table 2. The results for SPNv2 are taken directly from [9] since the augmentations are largely shared. The EKRN-HD, EKRN-HD-9, and EKRN-HD* make predictions with the same network (EKRN-HD), but the pose estimates are made with the EPnP algorithm, with the Top K strategy from [12], and with the DLSU algorithm respectively. The translation error (e_T) and rotation error (e_R) are reported as

$$e_T = \|\mathbf{t} - \mathbf{t}^*\|_2 \quad e_R = 2 \arccos \langle \mathbf{q}, \mathbf{q}^* \rangle \quad (14)$$

where $\mathbf{t}$ and $\mathbf{q}$ are the predicted translation vector and rotation quaternion and $\mathbf{t}^*$ and $\mathbf{q}^*$ are the respective true labels.

The accuracy comparison in Table 2 displays the results between networks with and without uncertainty predictions. The EKRN performs similarly on the test sunlamp and lightbox datasets compared to the KRN network. However, the EKRN is more accurate for the testing synthetic

dataset, which is of the same distribution that it trained on. This improvement may be caused by the benefits of multi-task learning, since the EKRN needs to not only predict the location of keypoints, but also their distribution.

By using only the top 9 keypoints with the highest confidence, the EKRN-HD-9 predictions across all testing datasets have slightly improved compared to using all keypoints in the EKRN-HD. Similarly, the EKRN-HD* uses the DLSU PnP method which utilizes the uncertainty estimates during pose estimation, allowing for increased accuracy across all datasets. This demonstrates the utility of predicting uncertainty, even if the uncertainty estimates are only used to improve the prediction accuracy. While robust estimation techniques like random sample consensus (RANSAC) [35] have been used in conjunction with PnP methods to remove outlier keypoints, uncertainty-aware PnP methods have been shown to further improve estimation accuracy [30]. A comparison of RANSAC approaches and uncertainty-aware PnP methods can be seen in Appendix A.

Finally, the EKRN-HD* network can be compared to the SPNv2 network. We see comparable translation predictions and improved rotation performance on the synthetic dataset, from the same distribution that the network was trained on, but slightly worse rotation performance on the out-of-distribution data while the translation errors are comparable or even better. The performance of the SPNv2 on the out-of-distribution datasets is likely caused by the multi-task learning strategy. Since the SPNv2 trains on additional information like the satellite segmentation maps, the multi-task framework is likely the cause for the better generalization to out-of-distribution data for orientation predictions.

According to the final results of the most recent satellite pose estimation competition in 2021 [23], the accuracy of the EKRN-HD* would have placed 5th in the lightbox category and 6th in the sunlamp category without viewing any sunlamp or lightbox images, unlike the other competitors who used online domain refinement techniques [9,25,24]. In future works, we aim to improve the generalization of the EKRN-HD network by performing multi-task learning, similarly to the SPNv2 net-

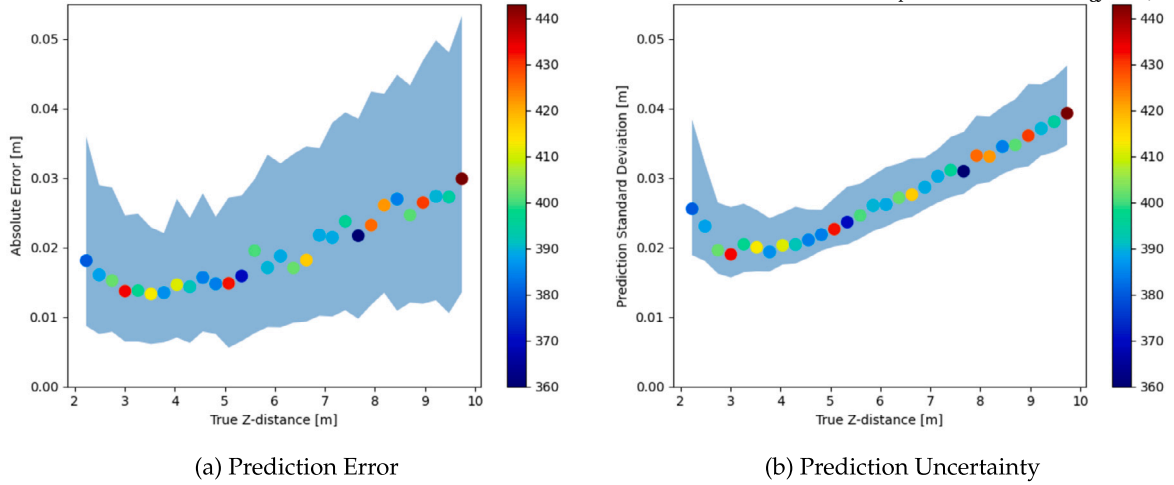


Fig. 5. The position error over distance from camera is compared in the **synthetic dataset**. The EKRN-HD-9 loses accuracy at further distances, which is predicted by the network's distribution uncertainty.

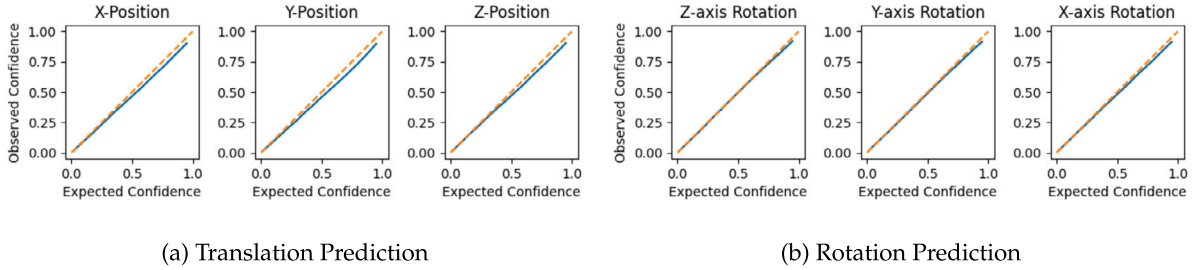


Fig. 6. Observed and expected confidence intervals are compared in the **synthetic dataset**. The EKRN-HD-9 accurately predicts the confidence intervals on the unseen testing data.

work, and apply online domain refinement using the out-of-distribution datasets.

According to [15], the EKRN network falls into a desirable regime for the balance between the number of parameters and regression error. Deployment on a satellite with limited hardware requires lighter networks with smaller memory and faster predictions. The EKRN network is comparable in size and accuracy to the KRN while additionally providing uncertainty estimates. The larger networks such as the EKRN-HD or SPNv2 may still be able to deploy onto satellites, but will require more memory and processing power for inference.

While the comparable accuracy of the EKRN-HD is notable, the main contribution of this work is the augmentation of uncertainty estimates. In the following figures, we focus on reporting the results for the EKRN-HD-9. While the EKRN-HD* produces the most accurate pose estimates, we are not able to randomly sample multiple poses since the DLSU algorithm requires the keypoint covariances. The EKRN-HD-9, however, is able to obtain pose samples since we can sample from the 9 most confident keypoints and use the EPnP algorithm to predict pose. The following results demonstrate the uncertainty quantification capability of the EKRN-HD-9 network.

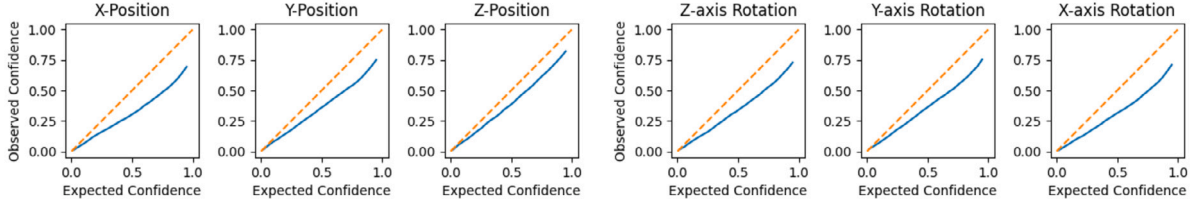
The distance of the satellite from the camera is largely dominated by the Z-axis distance in the camera frame. Fig. 5 displays the accuracy of the Z-axis distance prediction of EKRN-HD-9 over the actual Z-axis distance the satellite is away from the camera. The Z-axis distance prediction errors (y-axis) were separated into 30 bins across the true Z-axis distance (x-axis). The root median squared error (RMSE) was calculated for each bin and plotted with a colored point where the color denotes the number of test images in each bin. The shaded region denotes the 0.25-0.75 quantiles of the absolute error across the images in each bin. Since the RMSE increases as the distance increases, the network prediction is less accurate at further distances. The prediction uncertainty in

the Z-distance, also predicted by the network, accurately measures this error and follows the trend.

In Fig. 6, we compare the expected confidence interval with the observed confidence interval for the translation and rotation quantities of interest. For clarity, the rotation predictions are defined for a Z-Y-X rotation sequence which was converted from a predicted quaternion. For each testing image, 1,000 samples of the predictive distribution were taken and a range of quantiles were found from 0.025-0.975 of the empirical distribution. Across the testing dataset, the percent of times that the pose label was found inside each quantile interval is used as the observed confidence interval. For example, the observed confidence corresponding to the 95% expected confidence was calculated by finding the number of times the true pose label was found between the 0.025 and 0.975 quantiles of the predictive distribution. The expected and observed confidence intervals for the testing set of the synthetic data is nearly identical, with a very slight under-prediction of uncertainty. This result demonstrates the remarkable ability of the EKRN-HD-9 to predict uncertainty in the pose estimate while also providing similar accuracies to the SPNv2 network.

Figs. 7 and 8 display the observed and expected confidence of the evidential network across the sunlamp and lightbox datasets respectively. Since the lightbox and sunlamp data contain out-of-distribution images unseen by the network, the accuracy and predicted uncertainty is expected to be worse. The EKRN-HD-9 under-predicts the estimation uncertainty but is still able to capture most of the error. For example, the 95 expected confidence interval captures at worst 60 percent of the test images in the x-position and at best 80 percent in the z-position of the sunlamp dataset.

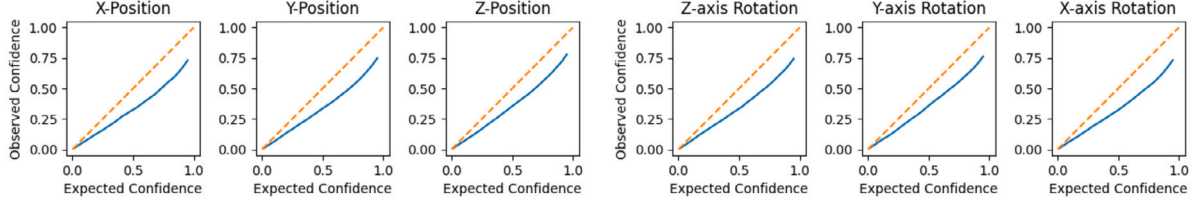
In Fig. 9, the absolute error between the mean prediction of the network (the γ -prediction) and the true label is seen as a function of the predicted uncertainty for the synthetic dataset. The absolute prediction



(a) Translation Prediction

(b) Rotation Prediction

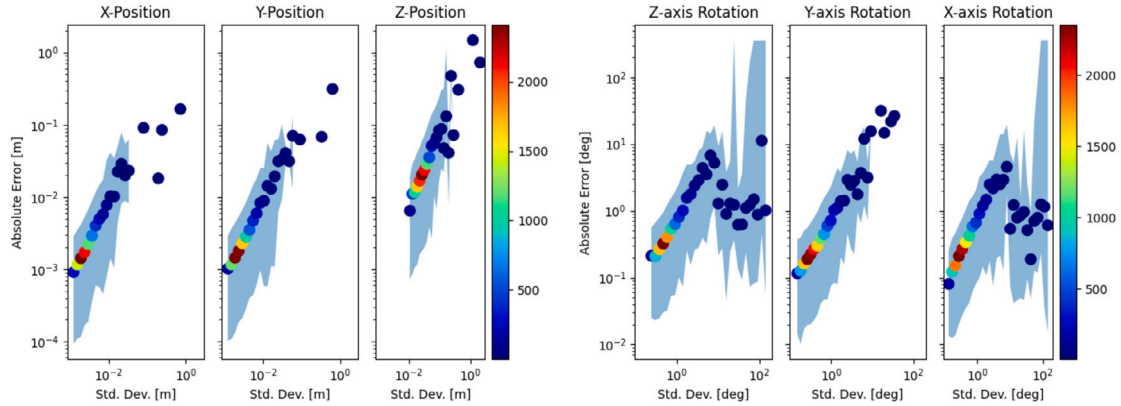
Fig. 7. Observed and expected confidence intervals are compared in the **sunlamp** dataset. The EKRN-HD-9 is able to capture most of the images in the predicted confidence interval.



(a) Translation Prediction

(b) Rotation Prediction

Fig. 8. Observed and expected confidence intervals are compared in the **lightbox** dataset. The EKRN-HD-9 is able to capture most of the images in the predicted confidence interval.



(a) Translation Prediction

(b) Rotation Prediction

Fig. 9. The RMSE of the prediction mean is compared to the prediction distribution's standard deviation in the **synthetic** dataset. The EKRN-HD-9 shows strong correlation between error and predicted uncertainty.

errors (y-axis) were separated into 30 bins across the predicted uncertainty (x-axis). The RMSE across the images in each bin was calculated and plotted with the colored points. The colors denote the number of test images in each bin and the shaded region denotes the 0.05-0.95 quantiles of the absolute error across the images in each bin. Notice that high predictive uncertainty corresponds to fewer images in each bin, creating a wider spread in the absolute error. In high-density areas of prediction, the RMSE correlates highly with the predicted uncertainty. Where there is lots of data, the EKRN-HD-9 predicts the uncertainty accurately and demonstrates a good correlation between error and predicted uncertainty. We also see that extremely large predicted uncertainty does not necessarily correspond to large errors.

Figs. 10 and 11 display the prediction error across prediction uncertainty for the sunlamp and lightbox datasets. While the prediction error is higher than the synthetic dataset, the network is able to predict higher uncertainty as well. Even though these datasets are on out-of-distribution images, there is still a strong correlation between predicted uncertainty and prediction error. These figures demonstrate the generalization of the EKRN-HD-9 to other satellite pose estimation domains.

We now aim to find the Pearson's correlation coefficient between the true error and predicted uncertainty for the EKRN-HD-9 network. To begin, the distributions of the 11 keypoints are predicted using the network with the testing datasets. The absolute errors between the means of the distribution (through γ -values) and the keypoints' labels are found. The standard deviations of the distributions are found using the total uncertainty in Equation (13). The correlation coefficients between this error and the standard deviations are recorded for each of the x-y coordinates of the 11 keypoints. The median across the 22 correlations are reported for each dataset in the first column of Table 3. The synthetic dataset has a median 0.996 correlation between the predicted uncertainty and the true error for the keypoint predictions. This demonstrates that the EKRN-HD network is able to accurately predict the uncertainty involved when predicting new keypoint locations. Similarly, when tested on out-of-distribution datasets, the EKRN-HD network is still able to report high correlations for the predicted keypoints.

To find the correlations when estimating pose, the keypoint distributions are sampled 1,000 times for each testing image. The pose is then predicted for each sample to create 1,000 positions and orientations for

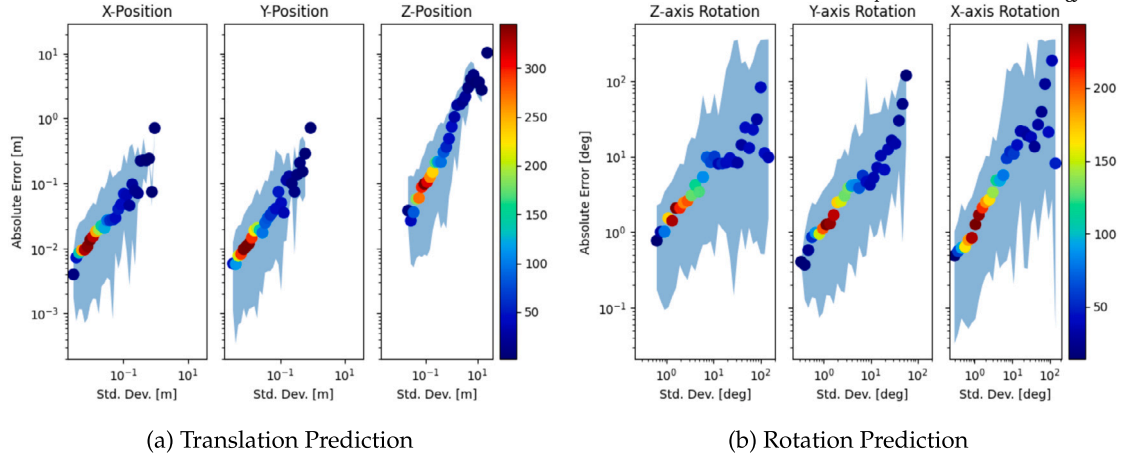


Fig. 10. The RMSE of the prediction mean is compared to the prediction distribution's standard deviation in the **sunlamp** dataset. The EKRN-HD-9 shows strong correlation between error and predicted uncertainty even in out-of-distribution images.

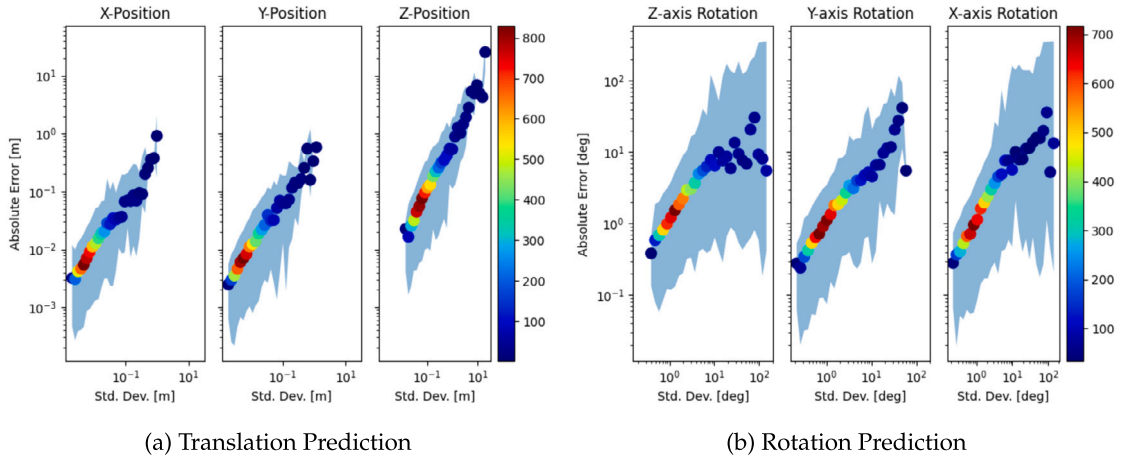


Fig. 11. The RMSE of the prediction mean is compared to the prediction distribution's standard deviation in the **lightbox** dataset. The EKRN-HD-9 shows strong correlation between error and predicted uncertainty even in out-of-distribution images.

Table 3

Median correlations between prediction error and predicted uncertainty for the EKRN-HD-9 network.

Dataset	Keypoint	Rotation	Translation
Synthetic	0.996	0.920	0.711
Sunlamp	0.553	0.583	0.895
Lightbox	0.806	0.563	0.927

one testing image. The error between the γ -mean of the distribution and the true label is recorded. To measure the spread of the position and orientation distributions, the distance of each sample from the γ -mean prediction is recorded for each of the 1,000 samples. The distance functions for the translation and orientation can be seen in Equation (14). This distance is then averaged across the samples for the orientation and translation and recorded for each image. Finally, the correlation across all testing images are found between the true error and the average distance of the translation and orientation samples. These correlations are reported in the second and third columns in Table 3. The high correlations for the synthetic dataset (0.996, 0.920, and 0.711) demonstrate the ability to predict the uncertainty of pose estimation. Similarly, the sunlamp and lightbox dataset correlations display the generalizability of the network to predict uncertainty on unseen data distributions.

Finally, we investigate the comparison between aleatoric and epistemic uncertainty in the predicted distributions of the keypoints. Fig. 12

shows one testing image with and without additive Gaussian noise. The EKRN-HD predicts the uncertainty for each keypoint and is able to separate epistemic and aleatoric uncertainty from Equation (13). The red crosses are the mean prediction points. Three standard deviations for the total uncertainty (lime green), epistemic uncertainty (blue), and aleatoric uncertainty (green) are shown in ellipses. The left image shows noise-free predictions with tight uncertainty bands. The right image, however, shows much larger uncertainty due to the additional noise. Most notably, the network identifies the epistemic uncertainty as the main contributor to the total uncertainty, while aleatoric uncertainty still increases, but is not the main contributor. While we do expect the aleatoric uncertainty to increase, since the noise that was added is inherently caused by the image, the epistemic uncertainty also increases. This indicates that the network would likely be able to further improve its accuracy by training over more images with this type of noise.

From the predictions that we have seen, most of the uncertainty predicted by the network is from epistemic uncertainty, as the network would likely be able to perform more accurately with more unique data. Future works will investigate the new utility of being able to identify the aleatoric and epistemic uncertainty in the training process.

6. Future work

The top-performing networks for the 2021 spacecraft pose estimation challenge tackled the inaccuracies due to the domain gap by including images of the lightbox and sunlamp datasets during training [23]. The

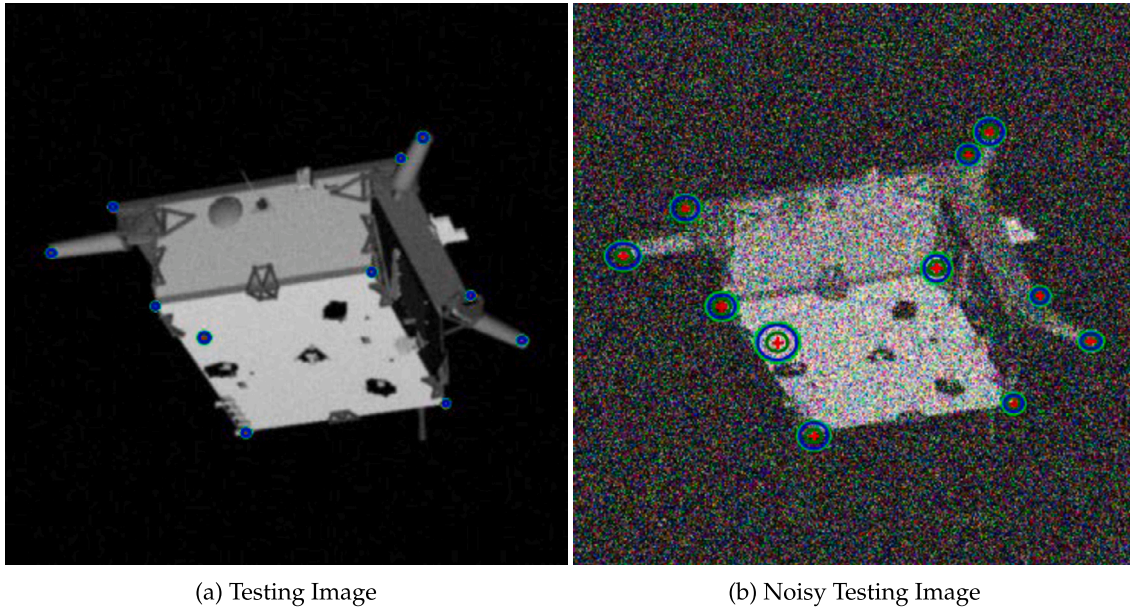


Fig. 12. A testing image is shown with keypoint predictions as red crosses, total uncertainty in lime green, epistemic uncertainty in blue, and aleatoric uncertainty in green. The EKRn-HD is able to separate uncertainty from a lack of data (epistemic) and from inherent data noise (aleatoric).

SPNv2 network was also able to further improve its performance using online domain refinement [9]. Since the evidential regression networks slightly underperform in accuracy on the out-of-distribution datasets, we aim to use similar methods in the future to close this gap.

Further, the evidential regression framework can be extended to other pose estimation networks by replacing the final network layer with four parallel layers to predict the four distribution hyperparameters. Extending the evidential framework specifically to the SPNv2 network would also require extending the loss function in Equation (12) to the multi-task learning objective function in [9].

Finally, uncertainty-aware pose predictions may be used in many decision-making algorithms. For example, many Kalman filtering algorithms require a measurement noise covariance term to track the trajectory of an object. Since our evidential network can predict pose uncertainty through a camera, we can inform the Kalman filter with a predicted covariance to track a satellite's trajectory more accurately. Learning the measurement noise for a Kalman filter has been shown to significantly impact its accuracy [36,37], and previous works have applied this to satellite pose estimation under uncertainty [10,11]. Another example is through trajectory optimization. If the observer needs to plan a route to or around the satellite, robust trajectory optimization algorithms require uncertainty information about their states [38,39]. By taking samples of the position and orientation of the satellite, a robust trajectory optimization algorithm would be able to find a safe and reliable trajectory for its mission. In future works, we aim to apply the EKRn-HD network to these tasks to demonstrate the importance of uncertainty quantification in decision-making processes.

7. Conclusion

In this work, we have presented a comprehensive approach to quantify uncertainty in satellite pose estimation using deep evidential regression. By leveraging both synthetic and testbed-created datasets, we demonstrated that our method effectively models uncertainties, providing reliable confidence measures of pose predictions. Our results indicate a strong correlation between the predicted uncertainty and the actual prediction error, affirming the robustness of our uncertainty quantification framework.

The integration of uncertainty estimates into the prediction process enhances decision-making by identifying when predictions are likely to

be less reliable. This capability is crucial for applications where the consequences of inaccurate pose estimation are significant, such as satellite docking and collision avoidance. Our approach not only maintains comparable accuracy with competitive models but also introduces a valuable layer of insight into prediction reliability.

Overall, our contributions advance the field of satellite navigation and pose estimation by introducing a novel approach to uncertainty quantification in deep learning. Future work will focus on improving the accuracy and uncertainty estimates while further reducing computational complexity, enabling real-time uncertainty estimation in operational scenarios.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. PnP method comparisons

We further investigate the utility of estimating keypoint uncertainty with respect to pose estimation accuracy. Without requiring keypoint uncertainty predictions, methods such as RANSAC repeatedly run a chosen PnP method to determine which keypoints worsen estimation accuracy. By iteratively testing random subsets of keypoints, RANSAC determines which keypoints are outliers and removes them from being used for estimation. In this section, we test and compare many PnP methods with and without RANSAC outlier rejection.

The optimal-PnP (OPnP) algorithm is considered to be a state-of-the-art PnP algorithm with high accuracy [40]. Compared to more popular approaches like EPnP, however, the OPnP algorithm earns higher accuracy at the expense of a higher computational cost [41,30]. Coupled

Table 4

Mean translation and rotation errors are reported for various PnP methods across all datasets with and without RANSAC preprocessing. Bold results indicate best in each column while green highlighted results indicate best in each dataset. The yellow highlighted PnP methods use feature uncertainty estimates.

	Synthetic				Sunlamp				Lightbox			
	RANSAC		All keypoints		RANSAC		All keypoints		RANSAC		All keypoints	
	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]	$\bar{e}_R$ [deg]	$\bar{e}_T$ [m]
EPnP [20]	0.0390	0.8247	0.0450	0.9014	1.1301	19.4719	1.1935	21.5230	0.6805	11.6618	0.7681	13.1538
DLS [43]	0.0367	0.7701	0.0364	0.7673	0.3468	13.2576	0.2836	15.2009	0.2540	8.2496	0.2216	8.6562
OPnP [40]	0.0371	0.7719	0.0375	0.7744	0.3585	12.8286	0.3633	14.8569	0.2892	8.1586	0.2641	8.5763
CEPPnP [27]	0.0427	1.0137	0.0408	0.9481	0.4022	17.7257	0.2843	17.1873	0.2355	9.8530	0.1953	9.4647
MLPnP [32]	0.0377	0.7524	0.0364	0.7546	1.2231	19.7059	1.2591	22.5048	0.7356	11.8773	0.7838	13.2769
DLSU [30]	0.0347	0.7890	0.0340	0.7503	0.3157	13.0376	0.2707	13.9545	0.2365	7.8608	0.1995	8.0121
DLSU* [30]	0.0364	0.8453	0.0358	0.8181	0.3157	13.0313	0.2215	13.8718	0.2241	8.0311	0.1829	8.0402
EPnP ^U [30]	0.0362	0.7871	0.0374	0.8087	1.1102	19.6136	1.1926	21.3286	0.6661	11.3797	0.7493	12.7292
EPnP ^U * [30]	0.0368	0.7959	0.0380	0.8150	1.0860	19.4984	1.2003	21.1542	0.6546	11.2239	0.7266	12.4010

with RANSAC, which evaluates a PnP algorithm repeatedly, we assume that a RANSAC-enabled OPnP algorithm will have higher estimation accuracy at a larger cost.

While OPnP may be considered the most accurate without using any feature uncertainty information, we may test many uncertainty-aware PnP methods against the RANSAC-enabled OPnP algorithm to determine if uncertainty values of keypoints improve estimation accuracy. While previous works have demonstrated the improved performance of uncertainty-aware PnP algorithms over traditional PnP approaches when all algorithms are fed RANSAC-filtered keypoints [30], we will test these methods on the predictions of the EKRN-HD network.

To find which keypoints are outliers for each test image, we use the RANSAC-enabled OPnP algorithm to find the subset of keypoints to throw out. With the outliers rejected, we pass the remaining keypoints to EPnP, DLS, OPnP, CEPPnP, MLPnP, DLSU, DLSU*, EPnP^U, and EPnP^U*. Although RANSAC may be used with any of the previous algorithms, we chose to only find the outlier keypoints with OPnP for its high estimation accuracy.

In Table 4, the average translation and rotation errors for various PnP methods are shown from the predictions of the EKRN-HD network across all testing datasets. Each method was tested either using all keypoints or only with the inlier keypoints determined by the RANSAC-OPnP preprocessing. The bold results demonstrate the best in the column while the green highlighted results report the best in each dataset. The PnP methods highlighted in yellow are the PnP methods which use the uncertainty estimates from the EKRN-HD predictions. The methods DLSU* and EPnP^U* require a prior pose estimate that may improve estimation. The prior pose estimate was created from the P3P method [42] with 3 randomly chosen keypoints.

The results shown in Table 4 were all created from the implementation of all the reported PnP methods in the Github repository <https://github.com/alexandervakhitov/uncertain-pnp> which was created for the paper that introduced DLSU and EPnP^U [30].

The DLSU and DLSU* methods contain 10 of the 12 best results by column. By dataset, the uncertainty-aware DLSU and DLSU* had 5 of the 6 lowest errors. The other two methods with a minimum error were the MLPnP method with the lowest RANSAC-synthetic translation error and the OPnP method with the lowest RANSAC-sunlamp translation error. Using RANSAC-enabled OPnP as a preprocessing tool to find the outlier keypoints did improve estimation accuracy for EPnP, OPnP, MLPnP, EPnP^U, and EPnP^U* across all datasets. The performance of DLS, DLSU, and DLSU*, however, was mixed across the datasets but generally seemed to favor using all keypoints. DLS, DLSU, and DLSU* had larger error for rotation and translation in the synthetic dataset when given the RANSAC preprocessed inliers. In the sunlamp and lightbox datasets, the translation errors were also worse when given RANSAC inliers. For the CEPPnP algorithm, all errors increased when given the inliers from RANSAC.

These results demonstrate that using RANSAC has a mixed effect across these PnP methods. The uncertainty-aware PnP methods, DLSU and DLSU*, however, generally outperformed PnP methods that did not include covariance feature information. By estimating uncertainty in keypoints using the EKRN-HD network, we allow the DLSU and DLSU* to have higher estimation accuracy compared to traditional PnP estimation methods.

Data availability

The SPEED+ dataset is publicly available at <https://purl.stanford.edu/wv398fc4383>.

References

- [1] Muzi Li, Bo Xu, Autonomous orbit and attitude determination for Earth satellites using images of regular-shaped ground objects, *Aerosp. Sci. Technol.* 80 (2018) 192–202.
- [2] Ben Guthrie, Minkwan Kim, Hodei Urrutxua, Jonathon Hare, Image-based attitude determination of co-orbiting satellites using deep learning technologies, *Aerosp. Sci. Technol.* 120 (2022) 107232.
- [3] Stefano Silvestrini, Margherita Piccinin, Giovanni Zanotti, Andrea Brandonisio, Ilaria Bloise, Lorenzo Feruglio, Paolo Lunghi, Michèle Lavagna, Mattia Varile, Optical navigation for lunar landing based on convolutional neural network crater detector, *Aerosp. Sci. Technol.* 123 (2022) 107503.
- [4] Benjamin B. Reed, Robert C. Smith, Bo J. Naasz, Joseph F. Pellegrino, Charles E. Bacon, The restore-1 servicing mission, in: *AIAA Space 2016*, 2016, p. 5478.
- [5] Jason L. Forshaw, Guglielmo S. Aglietti, Nimal Navarathinam, Haval Kadhém, Thierry Salmon, Aurélien Pisseloup, Eric Joffre, Thomas Chabot, Ingo Retat, Robert Axthelm, et al., Removedebris: an in-orbit active debris removal demonstration mission, *Acta Astronaut.* 127 (2016) 448–463.
- [6] Shengjie Wang, Yuxue Cao, Xiang Zheng, Tao Zhang, A learning system for motion planning of free-float dual-arm space manipulator towards non-cooperative object, *Aerosp. Sci. Technol.* 131 (2022) 107980.
- [7] Sumant Sharma, Connor Beierle, Simone D'Amico, Pose estimation for non-cooperative spacecraft rendezvous using convolutional neural networks, in: *IEEE Aerospace Conference*, 2018.
- [8] Tae Ha Park, Sumant Sharma, Simone D'Amico, Towards robust learning-based pose estimation of noncooperative spacecraft, *arXiv preprint, arXiv:1909.00392*, 2019.
- [9] Tae Ha Park, Simone D'Amico, Robust multi-task learning and online refinement for spacecraft pose estimation across domain gap, *Adv. Space Res.* 73 (2024) 5726–5740.
- [10] Lorenzo Pasqualetto Cassinis, Robert Fonod, Eberhard Gill, Ingo Ahrens, Jesús Gil-Fernández, Evaluation of tightly- and loosely-coupled approaches in cnn-based pose estimation systems for uncooperative spacecraft, *Acta Astronaut.* 182 (2021) 189–202.
- [11] Alexei Harvard, Vincenzo Capuano, Eugene Y. Shao, Soon-Jo Chung, Spacecraft pose estimation from monocular images using neural network based keypoints and visibility maps, in: *AIAA Scitech 2020 Forum*, 2020, p. 1874.
- [12] Keen Li, Haopeng Zhang, Chenyu Hu, Learning-based pose estimation of non-cooperative spacecrafts with uncertainty prediction, *Aerospace* 9 (10) (2022) 592.
- [13] Alexander Amini, Wilko Schwarting, Ava Soleimany, Daniela Rus, Deep evidential regression, *Adv. Neural Inf. Process. Syst.* 33 (2020) 14927–14937.
- [14] Tae Ha Park, Marcus Mörtens, Gurvan Lecuyer, Dario Izzo, Simone D'Amico, Speed+: next-generation dataset for spacecraft pose estimation across domain gap, in: *IEEE Aerospace Conference*, 2022.

- [15] Leo Pauly, Wassim Rharbaoui, Carl Shneider, Arunkumar Rathinam, Vincent Gaudilière, Djamila Aouada, A survey on deep learning-based monocular spacecraft pose estimation: current state, limitations and prospects, *Acta Astronaut.* 212 (2023) 339–360.
- [16] Yunfeng Zhang, Linjing You, Luyu Yang, Zhiwei Zhang, Xiangli Nie, Bo Zhang, Monocular 6-dof pose estimation of spacecrafts utilizing self-iterative optimization and motion consistency, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 6847–6856.
- [17] Michele Bechini, Geonmo Gu, Paolo Lunghi, Michèle Lavagna, Robust spacecraft relative pose estimation via cnn-aided line segments detection in monocular images, *Acta Astronaut.* 215 (2024) 20–43.
- [18] Pedro F. Proença, Yang Gao, Deep learning for spacecraft pose estimation from photorealistic rendering, in: *2020 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2020, pp. 6007–6013.
- [19] Julien Posso, Guy Bois, Yvon Savaria, Mobile-ursonet: an embeddable neural network for onboard spacecraft pose estimation, in: *2022 IEEE International Symposium on Circuits and Systems (ISCAS)*, IEEE, 2022, pp. 794–798.
- [20] Vincent Lepetit, Francesc Moreno-Noguer, Pascal Fua, Epnp: an accurate o(n) solution to the pnp problem, *Int. J. Comput. Vis.* 81 (2009) 155–166.
- [21] Sumant Sharma, Simone D'Amico, Neural network-based pose estimation for noncooperative spacecraft rendezvous, *IEEE Trans. Aerosp. Electron. Syst.* 56 (6) (2020) 4638–4658.
- [22] Mate Kisantal, Sumant Sharma, Tae Ha Park, Dario Izzo, Marcus Märtens, Simone D'Amico, Satellite pose estimation challenge: dataset, competition design, and results, *IEEE Trans. Aerosp. Electron. Syst.* 56 (5) (2020) 4083–4098.
- [23] Tae Ha Park, Marcus Märtens, Mohsi Jawaid, Zi Wang, Bo Chen, Tat-Jun Chin, Dario Izzo, Simone D'Amico, Satellite pose estimation competition 2021: results and analyses, *Acta Astronaut.* 204 (2023) 640–665.
- [24] Juan Ignacio Bravo Pérez-Villar, Álvaro García-Martín, Jesús Bescós, Marcos Escudero-Viñolo, Spacecraft pose estimation: robust 2d and 3d-structural losses and unsupervised domain adaptation by inter-model consensus, *IEEE Trans. Aerosp. Electron. Syst.* (2023).
- [25] Zi Wang, Minglin Chen, Yulan Guo, Zhang Li, Qifeng Yu, Bridging the domain gap in satellite pose estimation: a self-training approach based on geometrical constraints, *IEEE Trans. Aerosp. Electron. Syst.* (2023).
- [26] Alessandro Lotti, Dario Modenini, Paolo Tortora, Massimiliano Saponara, Maria A. Perino, Deep learning for real-time satellite pose estimation on tensor processing units, *J. Spacecr. Rockets* 60 (3) (2023) 1034–1038.
- [27] Luis Ferraz Colomina, Xavier Binefa, Francesc Moreno-Noguer, Leveraging feature uncertainty in the pnp problem, in: *Proceedings of the BMVC 2014 British Machine Vision Conference*, 2014, pp. 1–13.
- [28] Camille Chasset, Ron Nabeorn, Per Bodin, Robin Larsson, Björn Jakobsson, 3-axis magnetic control: flight results of the tango satellite in the prisma mission, *CEAS Space J.* 5 (1) (2013) 1–17.
- [29] Sumant Sharma, Tae Ha Park, Simone D'Amico, Spacecraft pose estimation dataset (speed), Stanford Digital Repository, 2019.
- [30] Alexander Vakhitov, Luis Ferraz, Antonio Agudo, Francesc Moreno-Noguer, Uncertainty-aware camera pose estimation from points and lines, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 4659–4668.
- [31] Bo Chen, Jiewei Cao, Alvaro Parra, Tat-Jun Chin, Satellite pose estimation with deep landmark regression and nonlinear pose refinement, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2019.
- [32] Steffen Urban, Jens Leitloff, Stefan Hinz, Mlpnp-a real-time maximum likelihood solution to the perspective-n-point problem, *arXiv preprint, arXiv:1607.08112*, 2016.
- [33] Alexander Buslaev, Vladimir I. Iglovikov, Eugene Khvedchenya, Alex Parinov, Mikhail Druzhinin, Alexandr A. Kalinin, Albumentations: fast and flexible image augmentations, *Information* 11 (2) (2020) 125.
- [34] Ilya Loshchilov, Frank Hutter, Decoupled weight decay regularization, *arXiv preprint, arXiv:1711.05101*, 2017.
- [35] Martin A. Fischler, Robert C. Bolles, Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography, *Commun. ACM* 24 (6) (1981) 381–395.
- [36] Vinay A. Bavdekar, Anjali P. Deshpande, Sachin C. Patwardhan, Identification of process and measurement noise covariance for state and parameter estimation using extended Kalman filter, *J. Process Control* 21 (4) (2011) 585–601.
- [37] Anirudh Chhabra, Jashwanth Rao Venepally, Donghoon Kim, Measurement noise covariance-adapting Kalman filters for varying sensor noise situations, *Sensors* 21 (24) (2021) 8304.
- [38] Thomas Lew, Riccardo Bonalli, Marco Pavone, Risk-averse trajectory optimization via sample average approximation, *IEEE Robot. Autom. Lett.* (2023).
- [39] Fenggang Wang, Shuxing Yang, FenFen Xiong, Qizhang Lin, Jianmei Song, Robust trajectory optimization using polynomial chaos and convex optimization, *Aerosp. Sci. Technol.* 92 (2019) 314–325.
- [40] Yingqiang Zheng, Yubin Kuang, Shigeki Sugimoto, Kalle Astrom, Masatoshi Okutomi, Revisiting the pnp problem: a fast, general and optimal solution, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2013, pp. 2344–2351.
- [41] Haoyin Zhou, Tao Zhang, Jayender Jagadeesan, Re-weighting and 1-point ransac-based pnp solution to handle outliers, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (12) (2018) 3022–3033.
- [42] Laurent Kneip, Davide Scaramuzza, Roland Siegwart, A novel parametrization of the perspective-three-point problem for a direct computation of absolute camera position and orientation, in: *CVPR 2011, IEEE*, 2011, pp. 2969–2976.
- [43] Joel A. Hesch, Stergios I. Roumeliotis, A direct least-squares (dls) method for pnp, in: *2011 International Conference on Computer Vision*, IEEE, 2011, pp. 383–390.